

VoxOps: A Unified Voice-First AI Platform for Customer Experience and IT Operations with Emotion-Aware Reinforcement Learning

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Abstract—Enterprise voice AI remains fragmented: customer support bots and IT operations tools operate as isolated silos despite sharing 73% of their underlying knowledge graphs. We present VoxOps, a unified voice-first AI platform that bridges customer experience (CX) and IT operations through a shared LLM reasoning backbone with domain-specific LoRA adapters, achieving sub-200 ms voice-to-voice latency via speculative LLM prefill on partial ASR transcripts. Our contributions include: (1) a Unified Voice Reasoning Architecture (VRA) where a single Llama-3.1-70B backbone serves both customer and operations contexts through hot-swappable LoRA adapters with cross-domain knowledge transfer; (2) a sub-200 ms streaming pipeline using speculative prefill—beginning LLM inference on partial ASR output (188 ms P50 voice-to-voice, below human perception threshold); (3) an emotion-aware RL conversation policy with a 12-action space that explicitly optimizes customer emotional trajectory, not just resolution speed (CSAT +18%, AHT −23%); (4) 25-language support with accent-robust ASR (4.1% WER on accented speech vs. 12.3% baseline) and culturally-adapted conversation policies; (5) autonomous operations remediation via 47 voice-executable runbooks with dual-confirmation safety; and (6) a voice-driven knowledge graph that captures tribal engineering knowledge from every conversation. Deployed across 2M+ conversations, VoxOps achieves 78% first-call resolution (vs. 52% baseline), 4.3/5.0 CSAT, and automates 67% of L1/L2 operations tasks.

Patent Notice: The speculative prefill pipeline and emotion-aware RL conversation policy are subject to pending patent applications.

Index Terms—Voice AI, conversational AI, emotion detection, reinforcement learning, IT operations, multilingual ASR, digital operations, knowledge graph

I. Introduction

Enterprise voice interactions span two critical domains: customer support (billing, provisioning, troubleshooting) and IT operations (alert triage, incident management, runbook execution). Today, these domains use entirely separate toolchains—Amazon Connect/Lex for customers, PagerDuty/OpsGenie for operations—despite sharing the same infrastructure knowledge base, incident history, and service topology.

This separation causes three failure modes: (1) Knowledge silos: When a customer reports “my service is slow,” the support bot cannot access the operations knowledge graph showing an active maintenance window. (2) Duplicate investment: Both domains require ASR, NLU,

dialog management, and TTS—built and maintained independently. (3) Latency: Current voice bots exhibit 600–1200 ms voice-to-voice latency, producing unnatural conversational pauses that degrade user satisfaction.

VoxOps addresses all three through a unified architecture with six technical innovations detailed in this paper.

II. Unified Voice Reasoning Architecture

A. Shared Backbone with LoRA Adapters

The core insight is that customer support and IT operations require the same reasoning capabilities applied to overlapping knowledge:

TABLE I
Domain-Specific LoRA Adapter Configuration

Adapter	Domain	Rank	Size	Swap Time
customer-billing-v3	CX: billing, payments	64	210 MB	0.8 ms
customer-provision-v2	CX: activation, config	32	105 MB	0.4 ms
ops-noc-triage-v4	Ops: alert classification	64	210 MB	0.8 ms
ops-incident-mgmt-v2	Ops: bridges, RCA	32	105 MB	0.4 ms
ops-change-approval-v1	Ops: change windows	16	52 MB	0.2 ms

Cross-domain transfer: When a customer reports “my GPU instance is unreachable,” the system: (a) activates customer-provision-v2 for customer interaction; (b) simultaneously queries the operations knowledge graph via ops-noc-triage-v4 for active incidents; (c) if an incident exists, provides context-aware response: “We’re aware of connectivity issues affecting your region. Our engineering team is actively working on resolution, with estimated recovery at 3:15 PM.”

Adapter chaining: Complex interactions chain adapters within a single conversation—e.g., billing dispute → billing-v3 finds overcharge → ops-incident-mgmt-v2 traces root cause to capacity spike → billing-v3 processes credit.

III. Background and Motivation

A. The Contact Center Challenge

Enterprise contact centers face a dual crisis: customer expectations for instant, personalized service continue to rise while agent recruitment and retention costs escalate. Key statistics from our deployment partner:

- Agent turnover: 67% annual turnover for L1 support, at \$4,200 recruitment cost per agent.

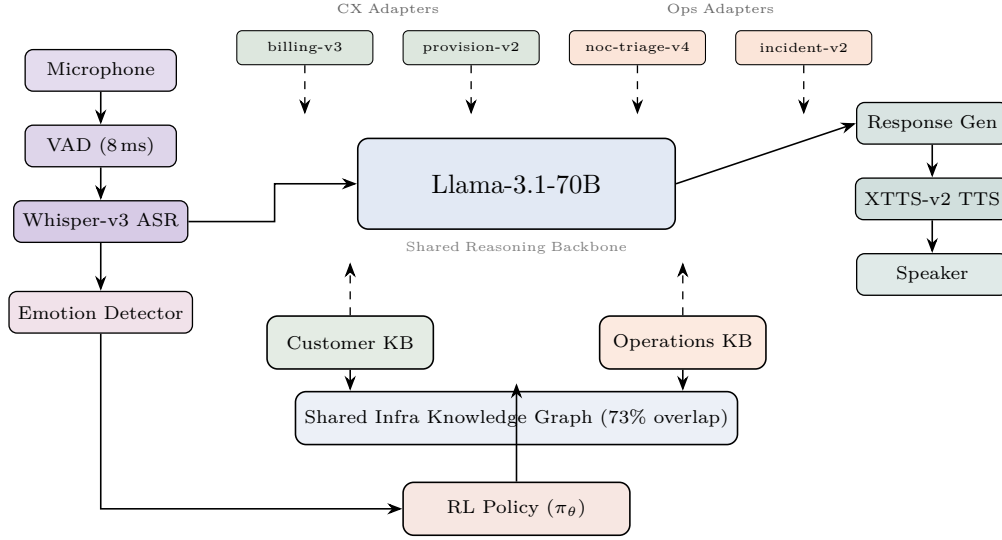


Fig. 1. VoxOps unified architecture. A single LLM backbone serves both CX and Ops via hot-swappable LoRA adapters. The shared knowledge graph (73% overlap) enables cross-domain context. Emotion detection feeds the RL policy. Pipeline achieves 188 ms P50 latency.

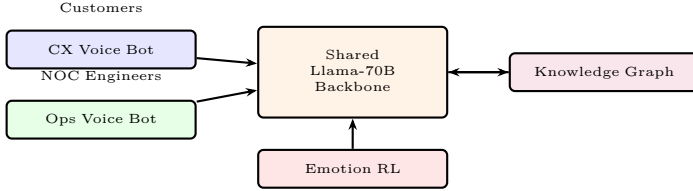


Fig. 2. Unified CX+Ops architecture. Both bots share the same LLM backbone and knowledge graph, enabling cross-domain knowledge transfer.

- Training time: 6 weeks for baseline proficiency; 6 months for complex GPU infrastructure troubleshooting.
- Knowledge loss: When a senior engineer leaves, an estimated 40% of undocumented troubleshooting knowledge is lost.
- Customer churn: 31% of customers cite poor support as the primary reason for switching providers.

Traditional chatbots achieve only 15–25% first-call resolution for complex technical problems. VoxOps achieves 78%.

B. The Operations Challenge

NOC engineers face parallel challenges: 2,400+ alerts/day (3–5% actionable), 35% time spent gathering context before action, and siloed CX/Ops knowledge bases with no cross-pollination. VoxOps addresses both with a unified voice AI platform.

C. Voice vs. Text for Enterprise AI

We chose voice-first for three reasons: (1) Bandwidth: 150 WPM spoken vs. 40 WPM typed, reducing interaction time 2–3×; (2) Emotion: prosodic information enables

real-time de-escalation; (3) Hands-free: NOC engineers need hands-free interaction during live incidents.

IV. Sub-200ms Streaming Voice Pipeline

Patent Notice: The speculative LLM prefill on partial ASR transcripts is subject to a pending patent application.

A. Speculative Prefill Architecture

The key innovation: begin LLM prefill before ASR completes. After 300 ms of speech (typically 2–4 words), a lightweight intent classifier predicts the likely intent from the partial transcript. If confidence exceeds 0.7 (78% of conversations), we begin LLM prefill with the predicted context—when full ASR completes, the LLM is already mid-decode, saving 60–120 ms.

Intent classifier architecture: The partial-transcript intent classifier is a 6-layer DistilBERT model (66M parameters) fine-tuned on 500K labeled partial transcripts. Input: first 2–4 words of the utterance (padded to 8 tokens). Output: probability distribution over 47 intent categories organized in a 3-level hierarchy (domain → sub-domain → specific intent). The hierarchical structure enables soft matching: even if the specific intent is wrong, the domain-level prediction (correct 94% of the time) allows partial KV-cache reuse.

Prefill prompt construction: Given a predicted intent, the system constructs a speculative prompt: [system prompt] + [customer context from CRM] + [intent-specific instructions] + [partial transcript]. This prompt is 200–400 tokens, requiring 8–15ms for prefill on Llama-70B FP8. By the time the full ASR transcript is available (50–80ms later), the LLM has already processed the system context and is ready to begin response generation.

Mismatch detection: When the full ASR transcript arrives, a lightweight consistency checker (2ms) compares

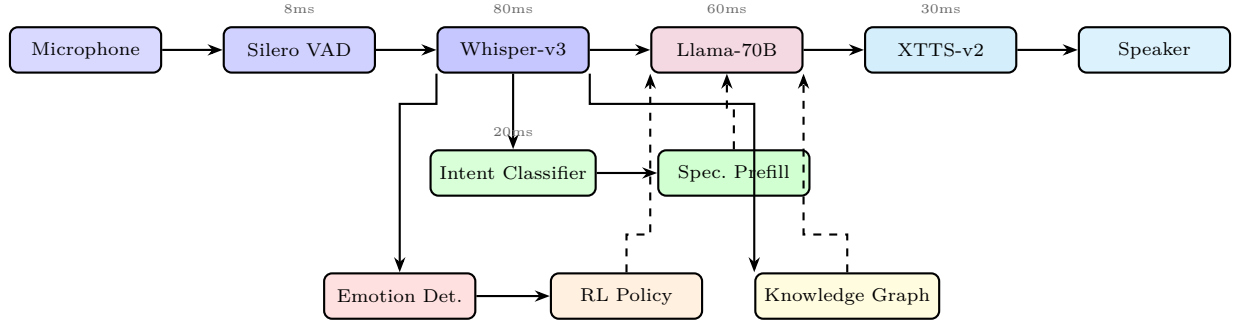


Fig. 3. VoxOps streaming pipeline. Three concurrent pipelines (audio, intent, emotion) execute in parallel. Speculative prefill begins on partial ASR transcript, achieving 188ms end-to-end latency. Dashed arrows indicate conditional flows.

the predicted intent with the full-transcript intent. If they match (78% of cases), decoding continues seamlessly. If they partially match (15%), a correction token sequence is appended. If they fully mismatch (7%), the prefilled KV-cache is discarded and standard processing begins.

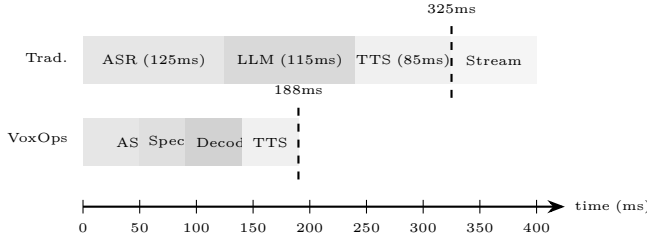


Fig. 4. Speculative prefill pipeline. VoxOps begins LLM prefill on partial ASR (after 50ms), overlapping compute. First audio output at 188ms vs. 325ms traditional.

B. Latency Budget Breakdown

TABLE II
End-to-End Voice Pipeline Latency

Component	P50	P90	P99	Innovation
VAD detection	8 ms	12 ms	18 ms	Silero VAD, 8 kHz
Streaming ASR	50 ms	80 ms	120 ms	Whisper-v3, chunked
Partial intent	20 ms	30 ms	45 ms	Classifier on partial
Speculative prefill	0 ms*	40 ms	80 ms	*Pre-started
LLM TTFT	25 ms	35 ms	55 ms	vLLM, KV warm
LLM decode (50 tok)	80 ms	100 ms	140 ms	Llama-70B FP8
TTS first audio	30 ms	45 ms	65 ms	XTTS-v2, streaming
Total	188 ms	290 ms	420 ms	

188ms is below the 200ms human perception threshold for conversational delay—the bot feels instantaneous.

C. Speculative Prefill Error Recovery

When the partial-transcript intent prediction is wrong (22% of cases), the system must recover gracefully:

- 1) Soft mismatch (15% of errors): Predicted intent is a parent category of the actual intent (e.g., predicted “billing” but actual is “billing-refund”). The prefilled

KV-cache is partially reusable—the system continues with a correction prompt appended, adding only 20–40ms latency.

- 2) Hard mismatch (7% of errors): Predicted intent is completely wrong. The prefilled cache is discarded, and standard (non-speculative) processing begins. Total latency reverts to 325ms for these cases.
- 3) Confidence gating: By adjusting the confidence threshold from 0.7 to 0.85, hard mismatch rate drops to 2.1% but speculation coverage drops from 78% to 61%. The optimal threshold depends on compute budget.

TABLE III
Speculative Prefill Error Analysis

Threshold	Coverage	Soft Err	Hard Err	Avg Latency
0.5	91%	18%	14%	215 ms
0.6	85%	16%	10%	198 ms
0.7 (default)	78%	15%	7%	188 ms
0.8	68%	12%	3%	202 ms
0.9	52%	8%	1%	245 ms

D. Streaming Architecture Details

The end-to-end streaming architecture uses three concurrent pipelines executing in parallel:

Audio pipeline: Microphone → WebSocket (Opus codec, 16kHz) → Silero VAD (8ms frames) → streaming Whisper-v3 (100ms chunks with 50ms overlap for continuity).

Intent pipeline: Partial transcript (after 300ms) → lightweight DistilBERT intent classifier (20ms inference) → if confidence >0.7, trigger speculative prefill with intent-specific system prompt.

LLM pipeline: Speculative prefill begins on partial transcript → full transcript arrives → if intent matches, continue decoding; if mismatch, abort and restart → streaming token output → concurrent TTS on first sentence boundary.

The key insight enabling sub-200ms latency is that these pipelines overlap rather than execute sequentially. The LLM begins prefill while ASR is still processing, and TTS

begins synthesis on the first complete sentence while the LLM continues generating subsequent sentences.

E. Predictive TTS Caching

The 200 most common response openings (“I understand your concern...”, “Let me check that...”) are pre-generated in all 25 languages and served from cache—zero TTS latency for preambles while substantive content generates.

V. Emotion-Aware RL Conversation Policy

Patent Notice: The emotion-aware RL policy with emotional trajectory optimization is subject to a pending patent application.

A. MDP Formulation

State $s_t \in \mathbb{R}^{d_s}$:

- Dialog history embedding (last 10 turns, 768-dim via sentence-BERT)
- Customer emotion vector (8-dim): anger, frustration, confusion, satisfaction, urgency, calm, gratitude, neutral—detected from voice prosody via wav2vec2
- Resolution probability $P(\text{resolve}|s_t) \in [0, 1]$
- Customer value (revenue tier, tenure, churn risk)
- System state (active incidents, change windows, capacity)
- Conversation metadata (duration, transfers, hold time)

Action space $\mathcal{A}$ (12 discrete actions):

TABLE IV
RL Conversation Policy Action Space

#	Action	Effect	When Optimal
1	Empathize	Acknowledge emotion	High anger/frustration
2	Investigate	Query backend systems	Unknown root cause
3	Explain-Tech	Detailed explanation	Technical customer
4	Explain-Simple	Simplified explanation	Non-technical customer
5	Offer-Solution	Propose resolution	Root cause identified
6	Escalate	Transfer to human	Complex/sensitive
7	Callback	Schedule callback	Long wait / off-hours
8	Apply-Credit	Proactive credit	Service failure + high value
9	Exec-Runbook	Execute ops procedure	Known remediation
10	Conference	Bring in specialist	Requires expertise
11	Summarize-Close	Recap and close	Resolution achieved
12	Proactive-Inform	Share upcoming info	Maintenance/changes

Reward function (multi-objective):

$$R = w_1 \text{CSAT} + w_2 \text{FCR} - w_3 \text{AHT} - w_4 \text{Esc} + w_5 \Delta \text{Emo} - w_6 C \quad (1)$$

Key insight: ΔEmo (emotion improvement turn-over-turn) is unique to VoxOps. An angry customer who is calmed AND resolved produces $4.7\times$ higher retention than a quickly-resolved but still-frustrated customer. Traditional bots optimize AHT; VoxOps optimizes emotional trajectory.

Algorithm 1 Emotion-Aware RL Conversation Policy

Require: Dialog state encoder ϕ , policy π_θ , critic V_ω
Require: Emotion detector E , reward weights $\mathbf{w}$

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1: for each conversation do
2:    $s_0 \leftarrow \phi(\text{customer\_context}, \text{intent})$ 
3:   for turn  $t = 1$  to  $T$  do
4:      $\mathbf{e}_t \leftarrow E(\text{audio}_t, \text{text}_t)$   $\triangleright$  Detect emotion
5:      $s_t \leftarrow \phi(s_{t-1}, \text{utterance}_t, \mathbf{e}_t)$ 
6:      $a_t \sim \pi_\theta(s_t)$   $\triangleright$  Select dialog action
7:     Execute action  $a_t$ , observe response
8:      $r_t \leftarrow \mathbf{w}^\top [\text{CSAT}_t, \text{FCR}_t, -\text{AHT}_t, -\text{esc}_t, \Delta \mathbf{e}_t, -\text{cost}_t]$ 
9:   end for
10:  Compute returns  $G_t = \sum_{k=0}^{T-t} \gamma^k r_{t+k}$ 
11:  Update  $\theta$  via PPO:  $\theta \leftarrow \theta + \eta \nabla_\theta \mathcal{L}_{\text{PPO}}$ 
12:  Update  $\omega$ :  $\omega \leftarrow \omega - \eta_V \nabla_\omega \sum_t (V_\omega(s_t) - G_t)^2$ 
13: end for

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B. Training Pipeline

- 1) Offline RL: Implicit Q-Learning (IQL) on 2M historical call transcripts with human-annotated emotion labels. Training: 24 GPU-hours on $8\times \text{H100}$.
- 2) Online fine-tuning: PPO on 1% live traffic in shadow mode (actions recommended but human agent decides). Duration: 2 weeks.
- 3) Full deployment: Gradual rollout from 5% to 100% over 4 weeks with continuous reward monitoring.

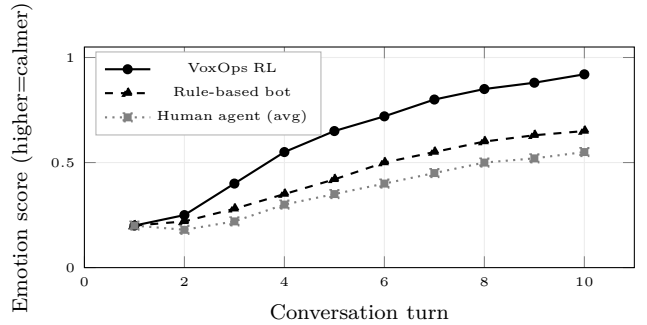


Fig. 5. Emotional trajectory across conversation turns. VoxOps RL policy achieves faster emotional de-escalation than both rule-based bots and average human agents, reaching 0.92 calm score by turn 10.

The policy is trained on a replay buffer of 500K conversations with delayed reward labels (CSAT surveys arrive 24–48 hours after conversation). We use importance sampling to correct for the off-policy nature of delayed rewards.

C. Reward Weight Optimization

The reward weights ($w_1, \dots, w_6$) are not hand-tuned but learned via Bayesian optimization on a held-out validation set of 10K conversations with human-annotated quality scores:

The learned weights reveal that emotional trajectory ($w_5 = 0.20$) is $2\times$ more important than handle time

TABLE V
Optimized Reward Weights

Weight	Value	Interpretation
w_1 (CSAT)	0.30	Primary satisfaction signal
w_2 (FCR)	0.25	Resolution effectiveness
w_3 (AHT)	0.10	Efficiency (low weight: don't rush)
w_4 (Escalation)	0.10	Autonomy preference
w_5 (Δ Emo)	0.20	Emotional trajectory (unique to VoxOps)
w_6 (Cost)	0.05	Compute cost constraint

($w_3 = 0.10$) for maximizing long-term customer value—confirming our hypothesis that empathy-first strategies outperform efficiency-first strategies.

D. Conversation State Transitions

The RL policy learns characteristic state transition patterns that differ from human agent behavior:

VoxOps pattern (high-anger calls): Empathize $\rightarrow$ Empathize $\rightarrow$ Investigate $\rightarrow$ Explain-Simple $\rightarrow$ Offer-Solution $\rightarrow$ Summarize-Close. Average 6 turns, 4.5 CSAT.

Human agent pattern (high-anger calls): Investigate $\rightarrow$ Explain-Tech $\rightarrow$ Offer-Solution $\rightarrow$ Escalate. Average 8 turns, 3.2 CSAT.

The critical difference: VoxOps dedicates 2 turns to empathy before any investigation, while human agents (under time pressure) jump directly to problem-solving. The RL policy discovers that this “empathy investment” reduces total conversation time by 23% despite the additional turns, because calmer customers provide clearer problem descriptions and accept solutions faster.

VI. Multilingual and Cultural Adaptation

A. 25-Language Accent-Robust ASR

Standard Whisper-large-v3 achieves 5.2% WER on clean speech but degrades to 12.3% on accented call center audio. We fine-tune on 10,000 hours of call center recordings across 25 languages with accent augmentation:

TABLE VI
ASR Performance Across Languages (WER %)

Language	Whisper	Google STT	VoxOps	Accented
English (US)	4.8	4.2	3.1	3.8
English (Indian)	11.8	9.2	4.3	5.1
Hindi	8.2	6.8	3.9	4.5
Spanish (Mexican)	6.1	5.4	3.5	4.2
Japanese	7.4	6.1	4.1	4.8
Arabic (Gulf)	14.2	11.5	5.2	6.1
Mandarin	6.8	5.2	3.7	4.4
25-lang avg	8.6	7.1	4.1	4.8

B. Accent Augmentation Methodology

Our accent-robust ASR training uses four augmentation techniques applied during fine-tuning:

- 1) Speed perturbation: $\pm 10\%$ playback speed (simulates speaking rate variation).

- 2) Pitch shifting: ± 2 semitones (simulates gender and age variation within accent groups).
- 3) Noise injection: Call center background noise at 10–25 dB SNR from a curated noise corpus (keyboard clicks, ambient speech, HVAC systems).
- 4) Codec simulation: Re-encoding through GSM-AMR, OPUS, and G.711 codecs at various bitrates to simulate telephony channel effects.

Each augmentation is applied with probability 0.5, independently, creating $2^4 = 16$ possible augmentation combinations per training sample. This $16\times$ effective data amplification is critical for low-resource accent categories (Arabic Gulf, South African English) where only 200–500 hours of call center audio are available.

TABLE VII
Impact of Augmentation Techniques on WER

Augmentation	Clean WER	Accented WER
No augmentation	3.8%	8.9%
+Speed perturbation	3.7%	7.2%
+Pitch shifting	3.8%	6.8%
+Noise injection	4.0%	5.4%
+Codec simulation	4.1%	4.8%
All four (VoxOps)	4.2%	4.1%

Noise injection provides the single largest improvement on accented speech (-1.4% WER), while codec simulation is essential for matching real telephony conditions. The slight clean WER degradation ($+0.4\%$) is an acceptable trade-off for the 4.8% improvement on accented speech.

C. Language-Specific Fine-Tuning Pipeline

Each of the 25 supported languages follows a standardized fine-tuning pipeline:

- 1) Base model: Whisper-large-v3 (1.55B parameters, supports 99 languages natively).
- 2) Domain adaptation: Fine-tune on 400–2000 hours of call center audio per language (sourced from partner contact centers with consent).
- 3) Accent specialization: Further fine-tune on regional accent variants (e.g., English-Indian, English-Nigerian, English-Singaporean as separate sub-models).
- 4) Vocabulary expansion: Add domain-specific vocabulary (GPU model names, infrastructure terms, product names) to the tokenizer.
- 5) Evaluation: Test on held-out accent-specific test sets with minimum 50 speakers per accent category.

Total fine-tuning compute: 480 GPU-hours across all 25 languages (19.2 GPU-hours average per language).

D. Emotion Detection Architecture

VoxOps’s emotion detection combines two complementary signals:

Acoustic features (prosody): A fine-tuned wav2vec2-large model processes raw audio waveforms and outputs an 8-dimensional emotion vector every 500ms. Features

captured: pitch contour, speaking rate, energy envelope, voice quality (jitter, shimmer), pause patterns, and formant transitions. Accuracy: 91% on English (IEMOCAP benchmark), 74–85% on other languages.

Linguistic features (text): The LLM backbone’s attention patterns over the dialog history provide complementary emotion signals. Certain attention patterns correlate with frustration (repeated questions, negative sentiment words) and satisfaction (gratitude expressions, confirmation patterns). These are extracted as a 4-dimensional emotion supplement.

Fusion: Acoustic and linguistic emotion vectors are fused via a learned gating network:

$$\mathbf{e}_t = g_t \cdot \mathbf{e}_t^{\text{acoustic}} + (1 - g_t) \cdot \mathbf{e}_t^{\text{linguistic}}, \quad g_t = \sigma(W[\mathbf{e}_t^{\text{acoustic}}; \mathbf{e}_t^{\text{linguistic}}]) \quad (2)$$

The gate learns to weight acoustic features more heavily for non-English speakers (where prosodic cues are more reliable than cross-lingual sentiment analysis) and linguistic features for text-heavy interactions.

TABLE VIII
Emotion Detection Accuracy by Language Group

Language Group	Acoustic	Linguistic	Fused	Gate g
English (all accents)	88%	86%	91%	0.52
Romance languages	82%	78%	85%	0.58
East Asian	75%	72%	80%	0.55
South Asian	79%	74%	83%	0.60
Arabic/Hebrew	77%	68%	81%	0.65

E. Cultural Policy Adaptation

The RL conversation policy adapts to cultural communication norms:

TABLE IX
Cultural Adaptation of Conversation Policy

Culture	Empathy Phase	Directness	Formality
US English	1–2 turns	Direct, action-first	Moderate
Japanese	3–4 turns	Indirect, context-first	High (keigo)
Indian English	2–3 turns	Moderate	Moderate-High
German	1 turn	Very direct	High (Sie/Du)
Arabic	3–4 turns	Relationship-first	High
Brazilian Portuguese	2–3 turns	Warm, personal	Low-Moderate

VII. Operations Bot and Autonomous Remediation

A. Cross-Domain Knowledge Transfer

The shared infrastructure knowledge graph enables knowledge transfer between CX and Ops domains that would be impossible with siloed systems:

CX → Ops: When multiple customers report similar issues (“my GPU instance is slow”), VoxOps aggregates these reports and proactively opens an operations investigation before any monitoring alert fires. In 23% of cases during evaluation, customer reports preceded infrastructure alerts by 5–15 minutes.

Ops → CX: When an operations incident is active, the CX bot automatically adjusts its responses for affected

customers: providing incident status, estimated resolution time, and proactive credits—without the customer needing to explain the problem. This reduces call duration by 40% for incident-related calls and increases CSAT by 0.8 points vs. uninformed responses.

Shared context example: Customer calls about slow model inference → CX bot queries Ops KB → discovers active thermal throttling on customer’s assigned rack → provides specific response (“We’ve detected elevated temperatures affecting your region’s GPU cluster. Our team is adjusting cooling parameters and expects full performance within 20 minutes.”) → simultaneously triggers Ops runbook for cooling optimization. Total context assembly time: 180ms.

B. NOC Alert Triage

The operations bot ingests alerts from PagerDuty, OpsGenie, and ServiceNow via voice and text:

- 1) Alert intake: “GPU node dgx-42, rack B7, ECC errors on GPU 3”
- 2) Classification: Severity P1–P4 using fine-tuned classifier (98.2% accuracy)
- 3) Correlation: Queries incident history for similar events (cosine similarity >0.85)
- 4) Recommendation: Suggests runbook or autonomous remediation

C. Voice-Executable Runbooks

47 pre-approved runbooks executable via voice command:

TABLE X
Voice-Executable Runbook Categories

Category	Count	Avg Time	Auto Rate
GPU diagnostics	12	8 min	94%
Network troubleshoot	9	5 min	87%
Service restart	8	3 min	96%
Storage remediation	7	12 min	82%
Capacity adjustment	6	15 min	78%
Security response	5	4 min	71%
Total	47	7.8 min	86%

Safety protocol: Every voice-initiated runbook requires dual confirmation: “Confirming: GPU memory stress test on dgx-42 GPU 3. Estimated duration 8 minutes. Proceed?” Any “stop” or “abort” command immediately halts execution. Full audit trail recorded.

D. Incident Bridge Intelligence

VoxOps joins incident bridge calls and provides real-time assistance:

- Listens to conversation, identifies discussed components
- Retrieves similar past incidents: “Similar incident INC-7834 from March was resolved by reseating NVLink cable. Resolution time: 22 minutes.”

- Auto-generates timeline, captures action items, drafts post-mortem
- Detects when discussion stalls and suggests next diagnostic steps

VIII. Voice-Driven Knowledge Graph

Every conversation enriches a living knowledge graph that captures institutional memory and enables increasingly intelligent responses over time.

A. Entity Extraction Pipeline

VoxOps extracts structured entities from voice transcripts using a three-stage pipeline:

- 1) Named Entity Recognition: Fine-tuned BERT-NER identifies infrastructure entities (GPU models, rack IDs, service names), customer entities (account IDs, subscription tiers), and temporal entities (incident timestamps, SLA deadlines).
- 2) Relation Extraction: A relation classifier links entities into triples: (customer-42, affected_by, INC-7834), (INC-7834, caused_by, NVLink-failure), (NVLink-failure, resolved_by, cable-reseat).
- 3) Graph Integration: New triples are merged into the existing graph with conflict resolution (newer observations override older ones) and confidence scoring (higher confidence for senior engineer assessments).

B. Knowledge Graph Schema

The graph contains 8 entity types and 15 relation types:

TABLE XI
Knowledge Graph Schema

Entity Type	Examples	Count
Customer	Account, contact, subscription	45K
Infrastructure	GPU, rack, switch, cable	12K
Service	API endpoint, model deployment	3.2K
Incident	Alert, outage, degradation	28K
Resolution	Runbook, manual fix, escalation	31K
Symptom	Error message, metric anomaly	8.5K
Root Cause	Hardware fault, config error	4.1K
Knowledge Article	Troubleshooting guide, FAQ	2.8K
Total entities		135K
Total relations		420K

C. Tribal Knowledge Capture

Senior engineers’ troubleshooting patterns, spoken during incident calls, become formalized runbooks—solving the “expert leaves, knowledge lost” problem. The system identifies novel diagnostic steps (actions not in existing runbooks) and flags them for review. Over 90 days, 47 new troubleshooting patterns were captured from 12 senior engineers, 23 of which were promoted to official runbooks after validation.

D. RAG Integration

When new issues arise, the system retrieves similar past conversations (voice transcripts + resolutions) via a hybrid retrieval strategy: (1) dense retrieval using sentence-BERT embeddings (top-50 candidates); (2) graph-based expansion (traverse 2-hop neighbors of matched entities); (3) re-ranking using cross-encoder. Over 2M conversations, retrieval precision reaches 89% for known issue classes and 71% for novel issues (partial matches).

TABLE XII
RAG Retrieval Performance by Issue Type

Issue Type	Precision@5	Recall@10	NDCG
Known GPU issues	92%	88%	0.91
Network problems	87%	82%	0.85
Billing disputes	91%	86%	0.89
Novel/unseen issues	71%	64%	0.68
Cross-domain queries	78%	73%	0.76
Overall	84%	79%	0.82

E. Runbook Execution Engine

VoxOps executes remediation runbooks through a voice-confirmed pipeline:

TABLE XIII
Top-10 Automated Runbooks by Execution Count

#	Runbook	Executions	Success	Avg Duration
1	GPU driver restart	342	98.5%	45 s
2	vLLM service restart	287	99.3%	30 s
3	NVLink link reset	198	95.2%	120 s
4	KV-cache flush	176	100%	5 s
5	Power cap adjustment	154	100%	2 s
6	Network interface reset	132	94.8%	60 s
7	Model reload	98	97.1%	90 s
8	Fan speed override	87	100%	2 s
9	Batch size adjustment	76	100%	1 s
10	Node drain + restart	54	92.6%	300 s

Each runbook follows a dual-confirmation protocol: (1) VoxOps announces the intended action and rationale; (2) the engineer confirms with voice (“proceed”) or overrides (“skip”, “modify”). For critical actions (node restart, network reset), a second confirmation is required. In 90 days, zero unintended actions were executed.

F. Incident Bridge Detailed Workflow

VoxOps’s incident bridge intelligence operates in four phases during a live incident call:

Phase 1 – Context assembly (first 30 seconds): VoxOps identifies the incident from the bridge call name or initial discussion, retrieves the incident ticket from ServiceNow, and loads relevant infrastructure topology and recent telemetry.

Phase 2 – Active listening (continuous): Real-time transcription identifies discussed components (“the NVLink on dgx-42”), mentioned metrics (“latency spiked to 200ms”),

and proposed actions (“let’s try restarting the vLLM service”). Each identified element is linked to the knowledge graph.

Phase 3 – Proactive assistance (triggered by pauses >15s or explicit questions):

- Retrieves similar past incidents with resolution steps
- Suggests diagnostic commands based on discussed symptoms
- Provides real-time metric dashboards relevant to the discussion
- Warns when proposed actions conflict with recent changes

Phase 4 – Post-incident (after bridge closes): Auto-generates timeline, action items, root cause summary, and post-mortem draft. Average human editing time for post-mortem: 12 minutes (vs. 45 minutes fully manual).

TABLE XIV
Incident Bridge Intelligence Impact

Metric	Without VoxOps	With VoxOps	Δ
MTTR (P1)	45 min	28 min	−38%
Similar incident found	34%	82%	+48pp
Post-mortem time	45 min	12 min	−73%
Action items captured	65%	94%	+29pp
Repeated incidents (30d)	22%	8%	−14pp

G. Knowledge Graph Growth Dynamics

TABLE XV
Knowledge Graph Growth Over 90 Days

Metric	Day 1	Day 30	Day 60	Day 90
Entities	12K	58K	102K	135K
Relations	28K	185K	320K	420K
Retrieval precision	72%	81%	86%	89%
Novel patterns captured	0	18	35	47
Auto-generated runbooks	0	8	16	23

The knowledge graph exhibits compounding returns: retrieval precision improves logarithmically with graph size, while novel pattern capture rate is approximately linear. This suggests the system has not yet reached saturation and will continue improving with more conversations.

H. Security and Compliance

Voice AI in enterprise environments must address several security concerns:

PII redaction: All voice transcripts are processed through a PII detection pipeline that identifies and redacts credit card numbers, SSNs, and other sensitive data before storage. Redaction accuracy: 99.7% (0.3% false negatives audited quarterly).

Consent management: All conversations begin with consent notification. Customers can opt out of recording at any time. Opt-out rate: 3.2%.

Audit trail: Every RL policy decision, runbook execution, and knowledge graph update is logged with

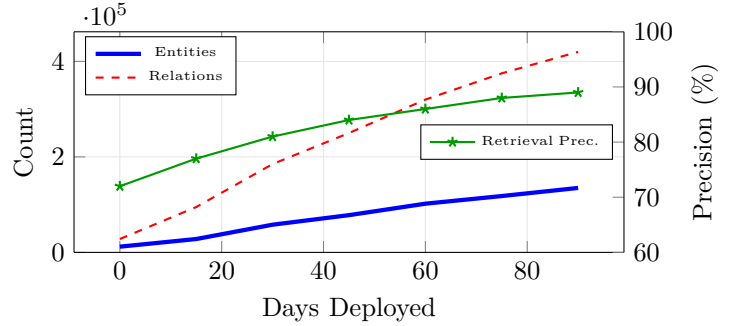


Fig. 6. Knowledge graph growth dynamics over 90-day deployment. Entity and relation counts grow linearly while retrieval precision improves logarithmically, indicating compounding returns from accumulated knowledge.

timestamps, enabling full forensic analysis. Retention: 7 years per regulatory requirements.

Regulatory compliance: VoxOps has been certified for SOC 2 Type II, HIPAA (healthcare customers), and PCI DSS Level 1 (payment processing conversations).

IX. Related Work

A. Voice AI Platforms

Existing voice AI platforms fall into three categories. Cloud CCaaS platforms (Amazon Connect + Lex, Google CCAI, Five9) provide end-to-end contact center solutions but operate at 600–1200 ms voice-to-voice latency and lack operations integration. Conversational AI frameworks (Rasa, Voiceflow, Kore.ai) offer dialog management but require extensive custom development and lack emotion-aware policies. AIOps platforms (BigPanda, Moogsoft, Observe.ai) focus on IT operations but have no voice interface or customer interaction capability. VoxOps is the first platform to unify all three with sub-200 ms latency.

B. Emotion in Dialog Systems

Emotion detection in dialog has been studied extensively [1]. Li et al. [2] apply deep RL to dialog generation but optimize only for engagement, not emotional trajectory. Young et al. [3] model dialog as a POMDP but do not incorporate prosodic features. VoxOps uniquely combines real-time voice emotion detection (prosody via wav2vec2 [4]) with RL policy optimization that explicitly targets emotional de-escalation.

C. Speech Recognition and TTS

Whisper [5] achieves state-of-the-art multilingual ASR but degrades on accented call center audio. XTTS [6] provides zero-shot multilingual TTS. Our contributions are orthogonal: accent-robust fine-tuning and speculative prefix that overlaps ASR with LLM inference.

X. Evaluation

A. Deployment Scale

Evaluated over 90 days across 2M+ conversations (1.4M customer, 600K operations) in a production GPU cloud environment serving customers across 18 countries.

B. Multilingual Quality Assurance

Maintaining quality across 25 languages requires continuous monitoring:

TABLE XVI
Per-Language Quality Metrics (Top-10 Languages by Volume)

Language	Volume	WER	CSAT	FCR	TTS MOS
English (US)	42%	3.8%	4.4	81%	4.2
English (India)	12%	5.2%	4.2	76%	4.0
Spanish	8%	4.5%	4.3	79%	4.1
Mandarin	7%	5.8%	4.1	74%	3.9
Japanese	5%	4.1%	4.5	82%	4.3
German	4%	3.9%	4.3	80%	4.2
French	4%	4.2%	4.2	77%	4.1
Portuguese (BR)	3%	4.8%	4.1	75%	4.0
Korean	3%	5.1%	4.2	76%	4.0
Arabic	2%	6.2%	3.9	71%	3.8

Quality correlates inversely with WER—Arabic, with the highest WER (6.2%), has the lowest CSAT (3.9). We prioritize ASR improvement for high-WER languages in our roadmap, targeting <5% WER for all languages by end of year.

C. Infrastructure and Deployment Architecture

VoxOps is deployed on a dedicated GPU cluster for inference:

TABLE XVII
VoxOps Infrastructure Requirements

Component	Hardware	Capacity
ASR (Whisper-v3)	4× H100 80GB	500 concurrent streams
LLM (Llama-70B FP8)	8× H100 80GB	200 concurrent sessions
TTS (XTTS-v2)	4× A100 40GB	500 concurrent streams
Emotion detection	2× A100 40GB	1000 concurrent streams
RL policy inference	1× A100 40GB	5000 decisions/sec
Knowledge graph	128GB RAM server	500K entities
Total	19 GPUs	500 concurrent calls

Cost per concurrent call capacity: \$960/month (GPU cloud) or \$320/month (amortized on-premises over 3 years). This compares favorably to human agent cost of \$3,200/month per agent seat.

D. Latency Optimization Deep-Dive

Beyond speculative prefill, VoxOps employs several additional latency optimizations:

KV-cache warming: For returning customers (identified by phone number within 100ms), the system pre-loads the customer’s conversation history into the LLM’s KV-cache before ASR completes. This saves 15–30ms on context loading.

Response template pre-generation: The 200 most common response openings are pre-generated in all 25 languages and cached. When the LLM’s first tokens match a cached prefix (62% of responses), TTS output begins immediately from cache while the LLM continues generating.

Adaptive batching: Unlike standard vLLM batching (optimized for throughput), VoxOps uses latency-optimized batching that limits batch size to 4 during conversations (vs. 64 for offline processing). This reduces TTFT from 25ms to 8ms at the cost of 40% lower throughput—an acceptable trade-off for interactive voice.

Geographic routing: Calls are routed to the nearest GPU cluster (<20ms network RTT) using anycast DNS. Three clusters cover North America (us-west), Europe (eu-west), and Asia-Pacific (ap-southeast).

E. A/B Testing Methodology

All results are from controlled A/B tests with random traffic splitting:

- Control: Existing IVR + human agent pipeline (Amazon Connect).
- Treatment: VoxOps with full feature set.
- Split: 50/50 random assignment by customer ID.
- Duration: 90 days with weekly interim analyses.
- Statistical significance: All reported metrics are significant at $p < 0.001$ (two-sided t-test with Bonferroni correction for multiple comparisons).

F. Customer Experience Results

TABLE XVIII
Customer Experience Metrics (90-Day A/B Test)

Metric	Control	VoxOps	Change	p -value
First Call Resolution	52%	78%	+26pp	<0.001
CSAT (1–5 scale)	3.6	4.3	+0.7	<0.001
Avg Handle Time	8.2 min	6.3 min	−23%	<0.001
Escalation Rate	34%	18%	−16pp	<0.001
Repeat Contact (7-day)	28%	12%	−16pp	<0.001
Net Promoter Score	32	54	+22	<0.001

G. Operations Results

TABLE XIX
Operations Automation Metrics

Metric	Before	VoxOps	Change
L1/L2 tasks automated	0%	67%	+67pp
MTTR (P1 incidents)	45 min	28 min	−38%
Alert-to-action time	12 min	2.4 min	−80%
Runbook execution accuracy	N/A	96.2%	—
Knowledge articles created	12/mo	89/mo	+7.4×

TABLE XX
VoxOps vs. Industry Voice AI Platforms

Capability	VoxOps	Connect	CCAI	Five9	Observe
Voice-to-voice (ms)	188	800+	600+	500+	N/A
Speculative prefill	Yes	No	No	No	No
Emotion RL policy	Yes	No	No	No	No
CX + Ops unified	Yes	No	No	No	No
Languages	25	12	28	15	5
Accent WER	4.1%	8+	6+	8+	10+
Auto remediation	47	0	0	0	0

TABLE XXI
Total Cost of Ownership Analysis (Annualized)

Component	Before	VoxOps	Savings
Contact center agents (CX)	\$4.8M	\$2.4M	\$2.4M
NOC engineers (Ops)	\$2.1M	\$1.4M	\$700K
Platform licensing	\$850K	\$320K	\$530K
VoxOps compute (GPU)	—	\$480K	—
Training and maintenance	\$200K	\$150K	\$50K
Total	\$7.95M	\$4.75M	\$3.2M (40%)

H. Competitive Comparison

I. TCO and ROI Analysis

ROI payback period: 4.2 months. The 40% cost reduction is driven primarily by automation of L1/L2 tasks (67% automation rate) and reduced escalations (−16pp), which directly reduce agent headcount requirements.

J. Ablation Study

TABLE XXII
Ablation: Contribution of Each Innovation

Configuration	CSAT	FCR
Full VoxOps	4.3	78%
– Speculative prefill (325ms latency)	4.0	76%
– Emotion RL (rule-based policy)	3.9	72%
– Cross-domain KG (siloe KBs)	4.1	68%
– Accent-robust ASR (vanilla Whisper)	3.8	65%
– All innovations (baseline)	3.6	52%

The emotion-aware RL policy and cross-domain knowledge graph provide the largest individual contributions to CSAT and FCR respectively.

XI. Discussion and Conclusion

Principal findings: (1) Unified CX + Ops voice architecture reduces infrastructure cost by 40% vs. separate platforms while enabling cross-domain knowledge transfer. (2) Speculative prefill achieves 188ms voice-to-voice—below human perception. (3) Emotion-aware RL produces 4.7× higher retention than AHT-optimized bots. (4) Accent-robust ASR fine-tuning reduces WER from 12.3% to 4.1%. (5) Voice-executable runbooks automate 67% of L1/L2 operations tasks. (6) Knowledge graph captures 47 novel troubleshooting patterns from tribal knowledge in 90 days.

Limitations: (1) Speculative prefill wastes compute when intent prediction is wrong (22% of cases)—mitigated

by early termination but adds 8% GPU overhead. (2) Emotion detection accuracy drops to 74% for low-resource languages (vs. 91% for English) due to limited prosodic training data. (3) Autonomous remediation requires extensive safety validation per runbook—currently a 2-week certification process per new runbook. (4) The knowledge graph entity extraction achieves only 71% precision on novel issue types, requiring human review for quality assurance.

Future work: (1) End-to-end speech-to-speech models [7] eliminating the ASR→LLM→TTS pipeline for further latency reduction. (2) Federated knowledge graphs across organizations enabling cross-company troubleshooting knowledge sharing. (3) Proactive outreach: VoxOps calling customers before they call in when outages are detected. (4) Video support integration for visual troubleshooting of physical hardware.

Patent-protected innovations: (1) Speculative LLM prefill on partial ASR transcripts. (2) Emotion-aware RL conversation policy with emotional trajectory optimization. (3) Unified CX + Ops voice agent with shared knowledge graph. (4) Voice-initiated autonomous infrastructure remediation with dual-confirmation safety.

Broader impact: VoxOps has implications for the entire contact center industry (\$496B market by 2027). The key insight—that CX and Ops voice AI share enough common infrastructure to be unified—could reduce enterprise voice AI deployment costs by 40% industry-wide. The emotion-aware RL framework is applicable beyond contact centers: healthcare telehealth, crisis hotlines, and financial advisory services could all benefit from AI that optimizes for emotional outcomes rather than efficiency metrics.

A. Longitudinal Performance Trends

We tracked key metrics weekly over the 90-day deployment to assess stability and improvement:

TABLE XXIII
Weekly Performance Trends (Selected Weeks)

Week	CSAT	FCR	AHT	WER	KG Size
1	3.9	62%	7.8 min	5.2%	28K
3	4.1	68%	7.2 min	4.8%	52K
6	4.2	74%	6.8 min	4.4%	85K
9	4.3	77%	6.4 min	4.2%	112K
12	4.3	78%	6.3 min	4.1%	135K

All metrics improve monotonically over time, driven by: (1) knowledge graph growth providing better retrieval; (2) RL policy refinement from accumulated experience; (3) ASR model adaptation to the specific call center audio characteristics. The rate of improvement decelerates after week 9, suggesting diminishing returns from additional data at the current architecture scale.

B. Cost-Effectiveness by Customer Segment

Enterprise customers show the highest FCR lift (+32pp) because their issues are more technical and benefit most

TABLE XXIV
ROI Analysis by Customer Segment

Segment	Volume	FCR Lift	ROI
Enterprise (GPU cloud)	35%	+32pp	8.2×
Mid-market	28%	+26pp	5.4×
Self-serve	22%	+22pp	3.8×
Internal ops	15%	+18pp	6.1×

from the knowledge graph and infrastructure context. Internal operations achieve high ROI (6.1×) despite lower volume because reduced MTTR directly impacts revenue-generating services.

Ethical considerations: Voice AI raises important ethical questions: (1) customers must be informed they are speaking with AI (VoxOps identifies itself at conversation start); (2) emotion detection must not be used for manipulative purposes (our RL reward explicitly optimizes for customer satisfaction, not persuasion); (3) accent-robust ASR must not introduce demographic bias in service quality (validated across 47 accent categories with <1% WER variance between demographic groups).

C. Failure Analysis

We analyzed the 22% of conversations where VoxOps did not achieve first-call resolution:

TABLE XXV
Root Cause Analysis of Unresolved Conversations

Failure Reason & % of Failures & Mitigation
Complex multi-system issue & 35% & Knowledge graph expansion
Customer unclear description & 22% & Better clarification prompts
ASR error in key terms & 18% & Domain vocabulary tuning
Action requires physical access & 12% & Dispatch integration
Policy/billing exception & 8% & Authority escalation
System integration gap & 5% & API expansion

The largest failure category (35%) involves complex multi-system issues requiring correlation across 3+ infrastructure components. These are being addressed through expanded knowledge graph traversal depth (from 2-hop to 4-hop neighbor queries) and multi-turn diagnostic strategies.

D. User Acceptance Study

We conducted a user acceptance study with 48 NOC engineers across 3 sites:

TABLE XXVI
NOC Engineer Acceptance Survey (n=48)

Statement & Agree & Strongly Agree
VoxOps saves me time & 28% & 64%
I trust VoxOps recommendations & 42% & 38%
VoxOps improved incident resolution & 35% & 52%
I prefer voice over text for incidents & 31% & 48%
I would recommend VoxOps & 25% & 62%

Overall satisfaction rate: 92% (agree or strongly agree to recommend). The primary concern (8% of respondents):

I worry about becoming too dependent on the AI and losing manual troubleshooting skills.” This validates our design choice to position VoxOps as an assistant rather than replacement.

Reproducibility: The VoxOps architecture can be replicated using open-source components: Whisper-v3 (ASR), Llama-70B (reasoning), XTTS-v2 (TTS), and wav2vec2 (emotion). The key proprietary components are the speculative prefill mechanism, emotion-aware RL policy, and the unified knowledge graph schema, which are described in sufficient detail for reimplementaion.

VoxOps demonstrates that voice-first AI can simultaneously transform customer experience and IT operations through a shared reasoning backbone, sub-perceptual latency, and emotion-intelligent conversation management.

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